English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

English transcription of the original handwritten solution outline. The original level of detail is retained, including the parts described there as standard.

Problem 1

1. Expected winnings

$$\mathbb{E}\left[\sum_{i=1}^{3}\left(\mathbf{1}_{\{\text{coin } i \text{ is heads}\}} + 3\mathbf{1}_{\{\text{die } i \text{ shows } 6\}}\right)\right] = 3\mathbb{P}(\text{coin 1 is heads}) + 3\cdot 3\mathbb{P}(\text{die 1 shows 6})$$
$$= \frac{3}{2} + \frac{9}{6} = 3.$$

2. A sum divisible by 6

Whatever the sum before round i, exactly one die outcome makes the new sum congruent to 0 modulo 6. Hence the number of rounds has distribution $\mathrm{Ge}(1/6)$. Put $p = 1/6$; then

$$\mathbb{E}[\text{winnings}] = \sum_{k=1}^{\infty} kp(1-p)^{k-1}$$
$$= \frac{p}{1-p}\sum_{k=1}^{\infty} k(1-p)^k = \frac{p}{1-p}\frac{1-p}{p^2} = \frac{1}{p} = 6.$$

This is precisely $\mathbb{E}[\mathrm{Ge}(1/6)]$.

3. Identifying a distribution

Since $\mathbb{P}(X=k)/\mathbb{P}(X=k-1) = 3/k$,

$$\mathbb{P}(X=k) = \frac{3}{k}\mathbb{P}(X=k-1) = \frac{3}{k}\frac{3}{k-1}\cdots\frac{3}{1}\mathbb{P}(X=0)$$
$$= \frac{3^k}{k!}\mathbb{P}(X=0).$$

Because the probabilities sum to 1,

$$\mathbb{P}(X=0)\underbrace{\left(1 + \frac{3}{1!} + \frac{3^2}{2!} + \cdots\right)}_{e^3} = 1.$$

Thus $\mathbb{P}(X=0) = e^{-3}$ and $\mathbb{P}(X=k) = e^{-3}3^k/k!$. Hence

$$X \sim \mathrm{Poi}(3), \qquad \mathbb{E}X = DX = 3.$$

4. Visitors at entrance 1

Let X be the number of visitors using entrance 1, and let $Y \sim \text{Poi}(\lambda)$ be the total number of visitors. Compute the generating function of X:

$$\begin{aligned}
\mathbb{E}s^X &= \sum_{k=0}^{\infty} \mathbb{E}[s^X \mid Y = k]\mathbb{P}(Y = k) \\
&= \sum_{k=0}^{\infty} \mathbb{E}\left[s^{\sum_{i=1}^{k} \mathbf{1}_{\{\text{visitor } i \text{ chooses entrance 1}\}}}\right] e^{-\lambda}\frac{\lambda^k}{k!} \\
&= \sum_{k=0}^{\infty} \left(\mathbb{E}s^{\mathbf{1}_{\{\text{visitor 1 chooses entrance 1}\}}}\right)^k e^{-\lambda}\frac{\lambda^k}{k!} \\
&= \sum_{k=0}^{\infty} \left(\frac{9}{10}s^0 + \frac{1}{10}s^1\right)^k e^{-\lambda}\frac{\lambda^k}{k!} \\
&= e^{-\lambda}e^{(9/10+s/10)\lambda} = e^{(\lambda/10)(s-1)}.
\end{aligned}$$

The third equality uses independence. For $Z \sim \text{Poi}(\lambda)$,

$$\mathbb{E}s^Z = \sum_{k=0}^{\infty} s^k e^{-\lambda}\frac{\lambda^k}{k!} = e^{-\lambda}e^{\lambda s} = e^{\lambda(s-1)}.$$

Comparing generating functions gives

$$X \sim \text{Poi}(\lambda/10), \qquad \mathbb{E}X = DX = \lambda/10.$$

In fact, as proved in the lectures, binomial thinning of $\text{Poi}(\lambda)$ with probability p has distribution $\text{Poi}(\lambda p)$.

Problem 2

1. Time to a winner

For $n \geq 3$,

$$\begin{aligned}
p_n := \mathbb{P}(\text{a winner exists}) &= \mathbb{P}(\text{one head, all others tails}) + \mathbb{P}(\text{one tail, all others heads}) \\
&= 2\frac{n}{2^n} = \frac{n}{2^{n-1}}.
\end{aligned}$$

Therefore $X \sim \text{Ge}(p_n)$, with

$$\mathbb{E}X = \frac{2^{n-1}}{n} = \frac{1}{p_n}, \qquad DX = \frac{1-p_n}{p_n^2}.$$

2. A fair entry price

The total payout is

$$100[(n-1) + (n-2) + \cdots + 2] = 100\left(\frac{(n-1)n}{2} - 1\right) =: S.$$

By symmetry, the expected winnings of player i do not depend on i. Thus

$$\sum_{i=1}^{n} \mathbb{E}[\text{winnings of player } i] = S = n\mathbb{E}[\text{winnings of player 1}].$$

The fair entry price is therefore S/n.

3. Total number of rounds

With k players remaining, part 1 gives an expected $2^{k-1}/k$ rounds until the next winner. Hence the expected total number of rounds is

$$\frac{2^{n-1}}{n} + \frac{2^{n-2}}{n-1} + \cdots + \frac{2^2}{3} = \sum_{k=3}^{n} \frac{2^{k-1}}{k}.$$

Problem 3

Parts 1, 2 and 3 are standard.

4. Set $Z_i = X_i - Y_i$. By the central limit theorem,

$$\begin{aligned}\mathbb{P}(X_1 + \cdots + X_n > Y_1 + \cdots + Y_n) &= \mathbb{P}(Z_1 + \cdots + Z_n > 0) \\ &\approx \mathbb{P}\left(N(0,1) > \frac{-n\mathbb{E}Z_1}{\sqrt{nDZ_1}}\right).\end{aligned}$$

The required mean and variance are

$$\mathbb{E}Z_1 = \mathbb{E}[X - Y] = \mathbb{E}X - \mathbb{E}Y,$$

using the quantities calculated in part 1, and

$$\begin{aligned}DZ_1 &= \mathbb{E}[(X-Y)^2] - (\mathbb{E}Z_1)^2 \\ &= \mathbb{E}X^2 + \mathbb{E}Y^2 - 2\mathbb{E}[XY] - (\mathbb{E}Z_1)^2.\end{aligned}$$

The second moments come from part 1, the product moment from part 3 (or directly), and the final squared mean from the preceding calculation.